

CM5235 - Homework Gaussian (2026)

Exercise 1 - Study of the ground state HCl diatomic molecule with different computational methods

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Single-point RHF/STO-3G at $R(\text{H-Cl}) = 1.275 \text{ \AA}$

SCF energy: $E(\text{RHF}) = -455.134835$ Hartree

Electric dipole moment: 1.7343 Debye (components $X = 0$, $Y = 0$, $Z = -1.7343$; along the bond axis)

Molecular orbitals: 10 in total = 9 occupied + 1 virtual. Orbital energies (Hartree):

MO	1	2	3	4	5	6	7	8 (HOMO)	9 (HOMO)	10 (LUMO)
E / Ha	-103.72252	-10.40401	-7.84887	-7.84216	-7.84216	-1.04362	-0.56448	-0.42315	-0.42315	+0.40398

Mulliken charges:

Atom	Cl	H
Charge	-0.171761	+0.171761

1) Optimisation + frequency with six basis sets

Route for each: #p hf/<basis> opt=tight freq.

Comparison table

Basis set	Optimal bond length / Å	Vib. frequency / cm ⁻¹	Number of orbitals	Number of basis functions	Total energy / Hartree	HOMO-LUMO gap / eV
STO-3G	1.3133	3372.5	10	10	-455.136012	21.72
6-31G	1.2953	2924.8	15	15	-460.037173	17.21
cc-pVDZ	1.2773	3149.2	23	23	-460.089450	16.87
6-311+G(d)	1.2714	3106.6	33	33	-460.087170	15.48
cc-pVTZ	1.2673	3138.3	48	48	-460.106851	16.37
cc-pVQZ	1.2656	3138.7	89	89	-460.111272	15.89
Experiment (NIST)	1.2746	2990.9 (ω_e)	-	-	-	-

Comparison with experimental values

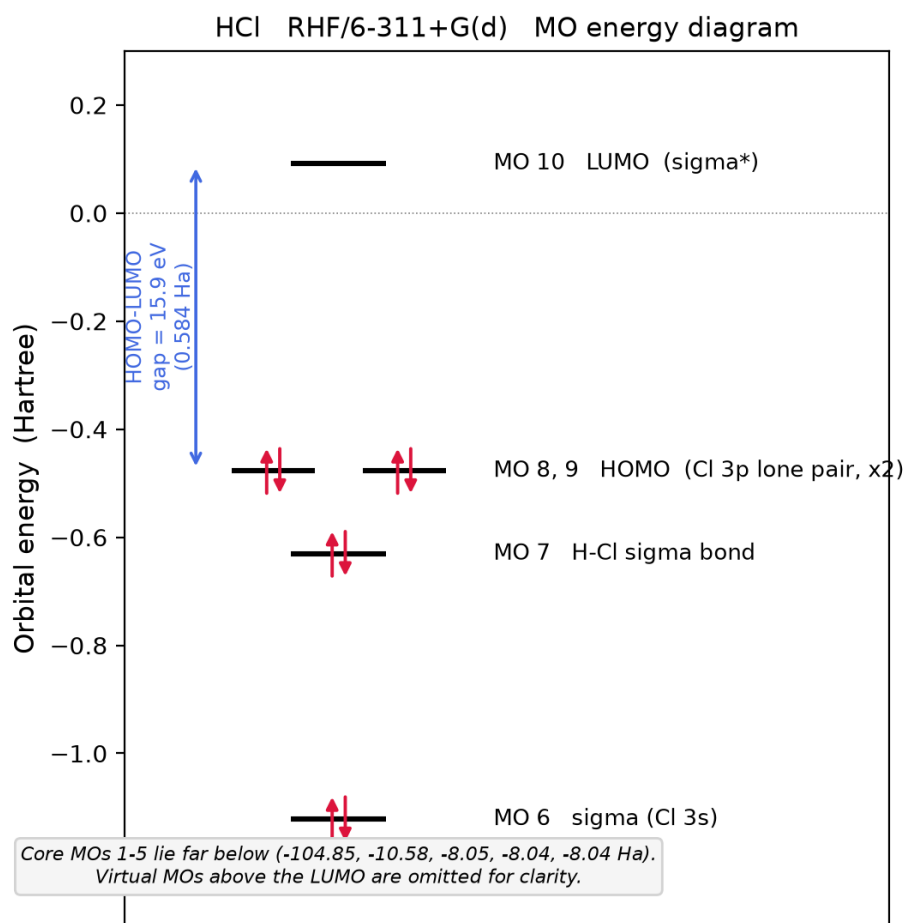
Experimental data from NIST CCCBDB (cccbdb.nist.gov/exp2x.asp, species HCl).

Quantity	STO-3G	6-31G	cc-pVDZ	6-311+G(d)	cc-pVTZ	cc-pVQZ	Experiment
Bond length R_e / Å	1.3133	1.2953	1.2773	1.2714	1.2673	1.2656	1.2746
Vibrational frequency / cm ⁻¹	3372.5	2924.8	3149.2	3106.6	3138.3	3138.7	2990.9
Moment of inertia / 1e-47 kg m ²	2.806	2.729	2.654	2.629	2.612	2.605	2.642
Zero-point energy / kJ mol ⁻¹	20.17	17.49	18.83	18.58	18.77	18.78	17.73
Enthalpy H(298)-H(0) / kJ mol ⁻¹	8.68	8.68	8.68	8.68	8.68	8.68	8.64
Entropy S(298) / cal mol ⁻¹ K ⁻¹	44.686	44.631	44.576	44.557	44.544	44.539	44.67

The calculated bond length, vibrational frequency and moment of inertia converge as the basis set is enlarged; at cc-pVQZ they differ from experiment by -0.009 Å, +148 cm⁻¹ and -0.037e-47 kg m² respectively, whereas the enthalpy H(298)-H(0) and the entropy agree with experiment to within 1%.

2) Orbital energy diagram and occupied MOs (RHF/6-311+G(d))

Orbital energy diagram



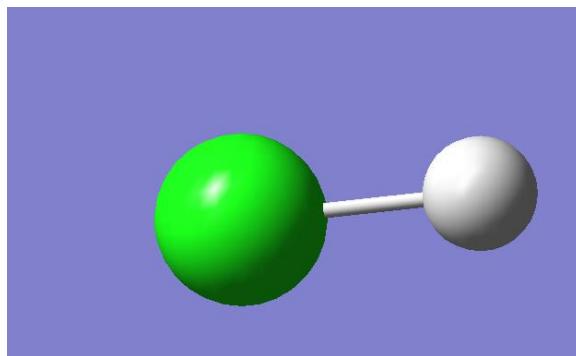
Orbital energy diagram of HCl, RHF/6-311+G(d). Core MOs 1-5 (-104.85 to -8.04 Ha) are off scale; virtual MOs above the LUMO are not shown.

MO	1	2	3	4	5	6	7	8 (HOMO)	9 (HOMO)	10 (LUMO)
E / Ha	-104.84972	-10.57650	-8.04538	-8.04148	-8.04148	-1.12190	-0.63065	-0.47730	-0.47730	+0.09172

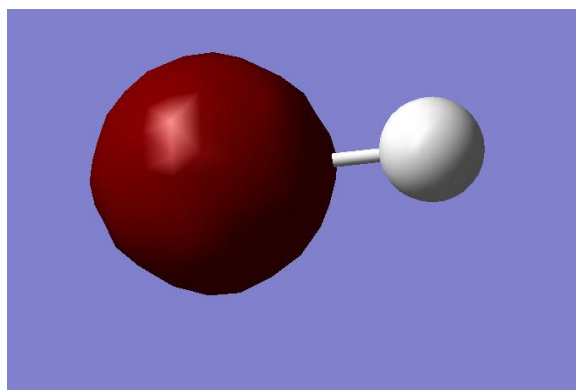
HOMO-LUMO gap = 0.09172 - (-0.47730) = 0.56902 Ha = 15.48 eV.

Occupied molecular orbitals (MO 1-9)

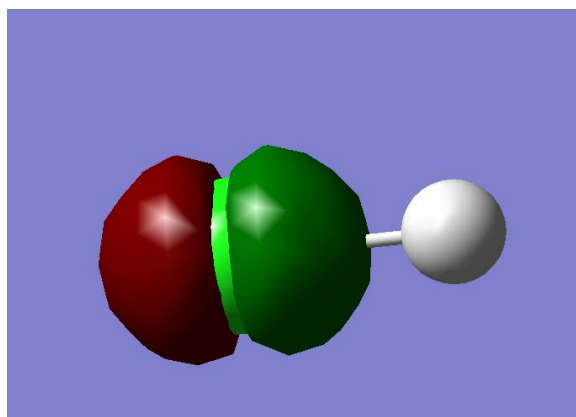
GaussView, isosurface value 0.02; red / green = the two phases of the wavefunction.



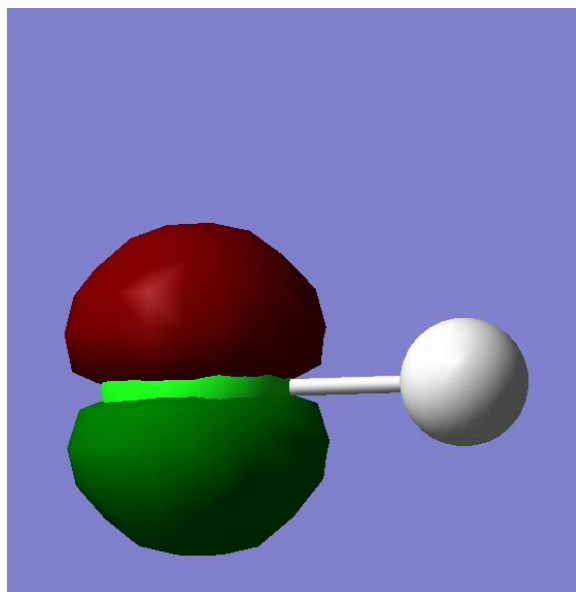
MO 1 ($E = -104.85$ Ha): Cl 1s core (too contracted to show an isosurface at 0.02).



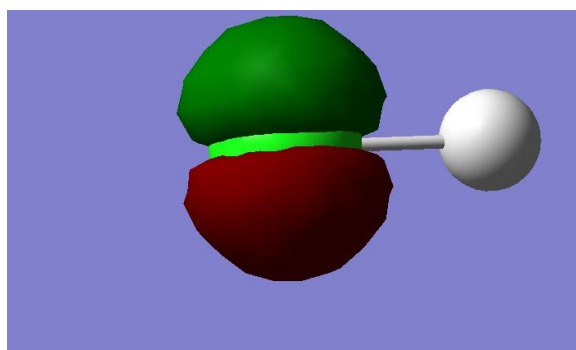
MO 2 ($E = -10.58$ Ha): Cl 2s core.



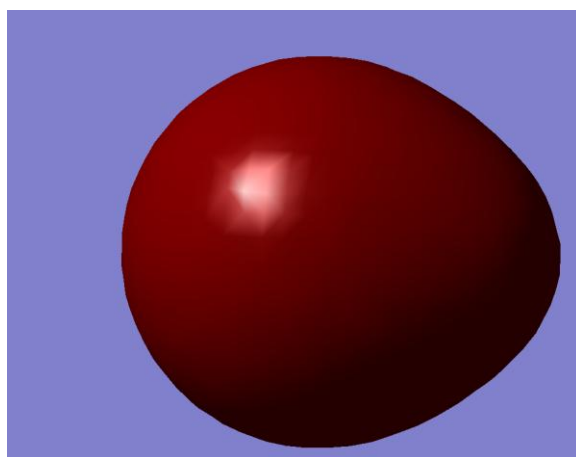
MO 3 ($E = -8.05$ Ha): Cl 2p core (along the bond axis).



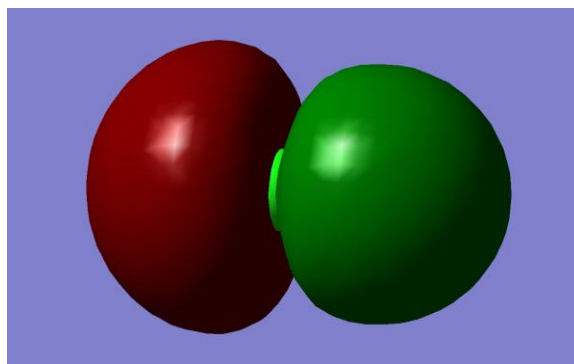
MO 4 ($E = -8.04$ Ha): Cl 2p core.



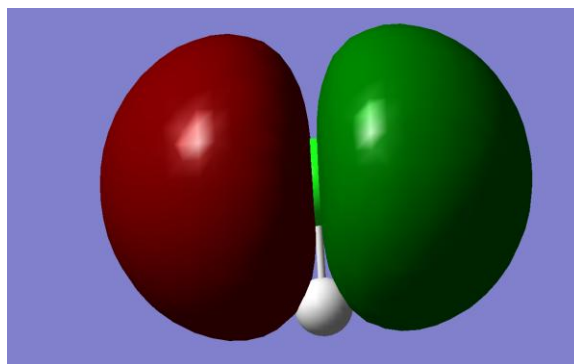
MO 5 ($E = -8.04$ Ha): Cl 2p core (degenerate with MO 4).



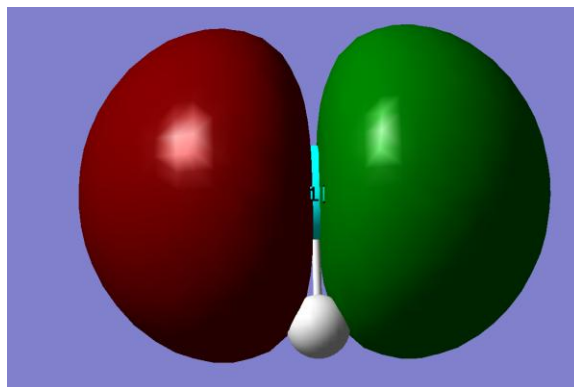
MO 6 ($E = -1.12$ Ha): sigma, dominated by Cl 3s.



MO 7 ($E = -0.63$ Ha): H-Cl sigma bond (Cl $3p_z$ + H $1s$).



MO 8 ($E = -0.48$ Ha): HOMO - Cl $3p$ lone pair, perpendicular to the bond.



MO 9 ($E = -0.48$ Ha): HOMO - Cl $3p$ lone pair, degenerate with MO 8.

Exercise 2 - Potential energy surface of the H-F molecule, four computational methods

a) Potential energy curves

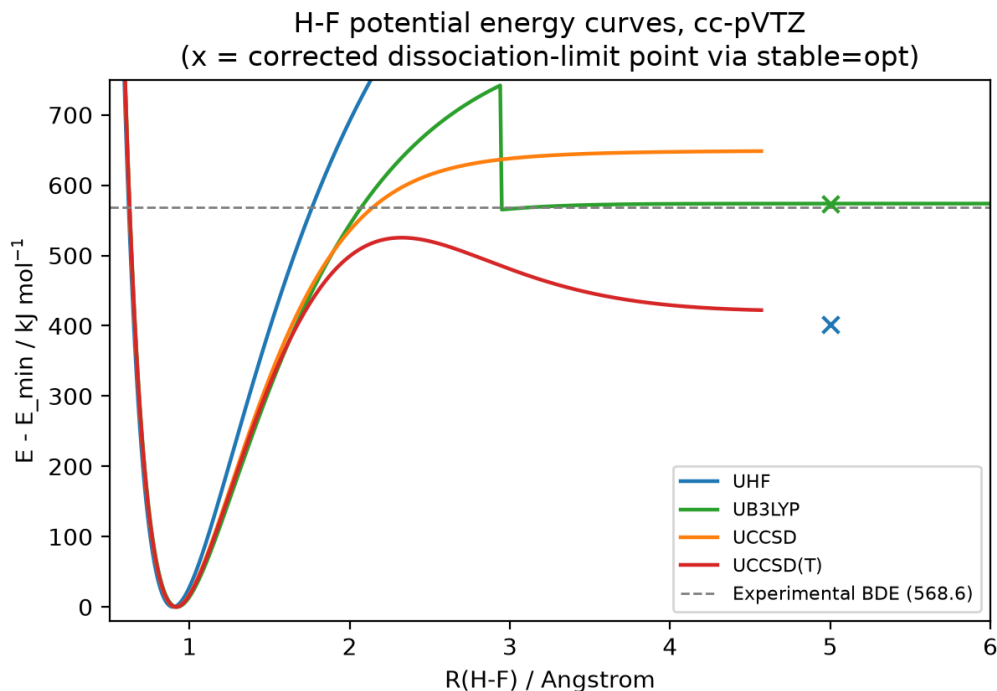


Figure 1. H-F potential energy curves, UHF/UB3LYP/UCCSD/UCCSD(T), cc-pVTZ.

b) Equilibrium bond length R_e

Method	$R_e / \text{Angstrom}$
UHF	0.90
UB3LYP	0.92
UCCSD	0.92
UCCSD(T)	0.92

c) Dissociation energy D_e

Method	$D_e / \text{kJ mol}^{-1}$
UHF	402.0
UB3LYP	573.8
UCCSD	648.4
UCCSD(T)	422.3

UCCSD and UCCSD(T) values are approximate: the amplitude equations failed to converge beyond $R = 4.57 \text{ \AA}$, so the last converged point was used in place of the true dissociation limit.

d) Comparison with the experimental bond dissociation energy (568.6 kJ/mol)

Method	D_e / kJ mol ⁻¹	Difference from experiment
UHF	402.0	-166.6
UB3LYP	573.8	+5.2
UCCSD	648.4	+79.8
UCCSD(T)	422.3	-146.3

UB3LYP is closest to experiment (off by only 5.2 kJ/mol) and is therefore the most accurate of the four methods here; UHF and UCCSD(T) underestimate D_e by over 140 kJ/mol and UCCSD overestimates it by 79.8 kJ/mol.

Exercise 3 - CH₂FCI bond dissociation energy

1) Total energy comparison, three methods

Method	Basis	E_total / Hartree
HF	STO-3G	-591.172393
B3LYP	cc-pVDZ	-599.362053
CCSD	cc-pVTZ	-598.670494

2) Electron density and electrostatic potential map (B3LYP/cc-pVDZ)

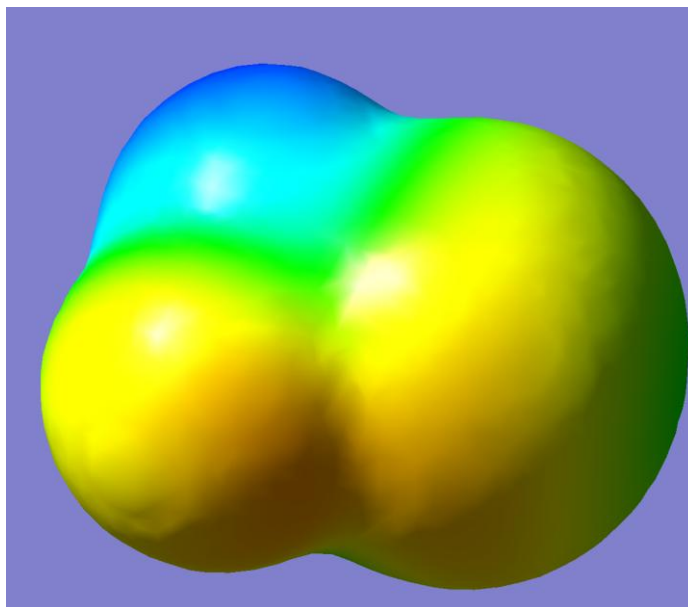


Figure 1. Total electron density surface (isovalue 0.0004) of CH₂FCI, mapped with the electrostatic potential (GaussView, B3LYP/cc-pVDZ). Blue = more positive potential, yellow/orange = more negative potential.

The surface itself is a total electron density isosurface (isovalue 0.0004), so its shape traces the outer envelope of the electron cloud: it bulges out furthest around Cl, which has the most electrons and the most diffuse (largest) electron cloud of the four elements present, and is noticeably tighter around the H atoms, which contribute only a single 1s electron each. Mapping the electrostatic potential onto this surface then shows where that density sits relative to the nuclear charges: the H atoms (blue) carry the most positive potential, while F and Cl (yellow/orange) are more negative, since both pull electron density toward themselves more strongly than C or H.

3) C-H and C-F bond dissociation energies

Strategy: homolytic bond cleavage; each radical/atom fragment is separately optimised (with frequencies, for ZPE) at UB3LYP/cc-pVDZ.

$$\text{BDE} = [\text{E}(\text{radical}) + \text{ZPE}(\text{radical}) + \text{E}(\text{atom})] - [\text{E}(\text{parent}) + \text{ZPE}(\text{parent})]$$

C-H bond, B3LYP/cc-pVDZ: CH₂FCl -> CHFCl. + H.

Species	E / Hartree	ZPE / Hartree
CH ₂ FCl (parent)	-599.362053	0.030710
CHFCl. radical	-598.698761	0.017241
H atom	-0.501258	0 (atom, no vibration)

BDE(C-H) = 0.148565 Hartree = 390.1 kJ/mol

C-F bond, B3LYP/cc-pVDZ: CH₂FCl -> CH₂Cl. + F.

Species	E / Hartree	ZPE / Hartree
CH ₂ FCl (parent)	-599.362053	0.030710
CH ₂ Cl. radical	-499.461337	0.022429
F atom	-99.726601	0 (atom, no vibration)

BDE(C-F) = 0.165834 Hartree = 435.2 kJ/mol

The C-F bond is the stronger of the two (435.2 vs 390.1 kJ/mol); the H atom is the easier of the two to remove from CH₂FCl.

Exercises 4 and 5 - Theoretical questions

Exercise 4 - Theoretical question 1

Describe the basic approximations used in the Hartree-Fock Self Consistent Field Method: Born-Oppenheimer Approximation, Orbital Approximation, Mean Field Approximation. How do these approximations simplify the computational problem?

Born-Oppenheimer. Nuclei are about 1836 times heavier than an electron, so on the timescale of nuclear motion the electrons relax essentially instantly to their ground state at any fixed nuclear geometry. The total wavefunction is then separated as $\Psi = \Psi_{\text{el}}(\mathbf{r}; \mathbf{R}) \cdot \chi_{\text{nuc}}(\mathbf{R})$, with $\mathbf{R}$ entering the electronic problem only as a parameter. Solving $\hat{H}_{\text{el}}\Psi_{\text{el}} = E_{\text{el}}(\mathbf{R})\Psi_{\text{el}}$ once per fixed geometry, instead of together with the nuclei, turns one coupled problem into a 3N-electron electronic problem plus a potential energy surface $E_{\text{el}}(\mathbf{R}) + V_{\text{nn}}(\mathbf{R})$ that can be optimised, scanned or differentiated afterwards.

Orbital approximation. The N-electron wavefunction is approximated by a single Slater determinant of N one-electron spin orbitals, $\Psi = (1/\sqrt{N!}) \cdot \det[\chi_1(x_1) \dots \chi_N(x_1); \dots; \chi_1(x_N) \dots \chi_N(x_N)]$, which builds in the Pauli principle automatically (swapping two electrons flips the sign; two electrons in the same spin orbital give a zero determinant). One unknown function of 3N coordinates becomes N unknown one-electron functions of 3 coordinates each, and each of those can be expanded in a finite atomic-orbital basis - so finding the orbitals is now a linear-algebra (matrix diagonalisation) problem rather than a differential equation.

Mean-field approximation. Each electron is treated as moving in the time-averaged charge distribution of the others rather than reacting to their instantaneous positions. This turns the electron-electron repulsion into an effective one-electron Fock operator, $f(1) = h(1) + \sum_j [J_j(1) - K_j(1)]$, $f\phi_i = \epsilon_i\phi_i$, so the N-electron problem is replaced by N one-electron eigenvalue problems, solved iteratively until self-consistent (SCF) because f depends on the orbitals it produces. This is far cheaper than the exact many-body problem, at the price of leaving out electron correlation - the instantaneous part of the electron-electron interaction.

Exercise 5 - Theoretical question 2

Describe the covalent chemical bonding on the example of the H₂ molecule. Write down the Slater Determinant for the ground state and give an estimate of the total energy using the method explained in Lecture 3 (with h , J , K). Write down the energies of all excited Slater Determinants that can be formed using the minimal basis set on each of the H atoms. Explain the Coulomb and Exchange integrals.

Bonding picture. Each H atom contributes one 1s orbital, a on atom A and b on atom B, with overlap $S = \langle a | b \rangle$. Symmetry gives a bonding and an antibonding combination:

$$\sigma_g = \frac{a + b}{\sqrt{2(1 + S)}} \quad \sigma_u = \frac{a - b}{\sqrt{2(1 - S)}}$$

σ_g piles electron density up between the nuclei (lowering the energy - the covalent bond); σ_u has a node there and lies higher in energy.

Ground-state determinant and energy. Both electrons occupy σ_g with opposite spin:

$$\psi_0 = \frac{|\sigma_g(\alpha) \sigma_g(\beta)|}{\sqrt{2}}$$

With $E = \sum_i h_{ii} + \sum_{i < j} (J_{ij} - K_{ij}) + V_{nn}$ and only one opposite-spin pair ($K = 0$):

$$E_0 = 2h_{gg} + J_{gg} + V_{nn}$$

$h_{gg} = (h_{aa} + h_{ab})/(1+S)$ and J_{gg} is the Coulomb self-repulsion of the σ_g charge cloud. The resonance integral $h_{ab} = \langle a | \hat{h} | b \rangle$ is negative and sizeable - it is what pulls E_0 below the energy of two separated atoms ($2h_{aa}$) and gives the covalent bond.

Excited determinants. With 2 spatial orbitals (σ_g, σ_u) there are 4 spin orbitals, so besides the ground configuration there are 5 more determinants:

Determinant	M_S	Energy
$\sigma_g(\alpha) \sigma_g(\beta)$ (ground state)	0	$2h_{gg} + J_{gg} + V_{nn}$
$\sigma_g(\alpha) \sigma_u(\alpha)$	+1	$h_{gg} + h_{uu} + J_{gu} - K_{gu} + V_{nn}$
$\sigma_g(\beta) \sigma_u(\beta)$	-1	$h_{gg} + h_{uu} + J_{gu} - K_{gu} + V_{nn}$
$\sigma_g(\alpha) \sigma_u(\beta)$	0	$h_{gg} + h_{uu} + J_{gu} + V_{nn}$
$\sigma_g(\beta) \sigma_u(\alpha)$	0	$h_{gg} + h_{uu} + J_{gu} + V_{nn}$
$\sigma_u(\alpha) \sigma_u(\beta)$ (doubly excited)	0	$2h_{uu} + J_{uu} + V_{nn}$

The two $M_S = 0$ singly-excited determinants are not eigenstates by themselves; their symmetric and antisymmetric combinations give the spin-adapted triplet (energy $h_{gg} + h_{uu} + J_{gu} - K_{gu} + V_{nn}$) and singlet ($h_{gg} + h_{uu} + J_{gu} + K_{gu} + V_{nn}$) excited states.

Coulomb and exchange integrals. Coulomb integral:

$$J_{ij} = (ii | jj) = \iint \phi_i(1)^2 (1/r_{12}) \phi_j(2)^2 d\tau$$

This is the classical electrostatic repulsion between the charge clouds $|\phi_i|^2$ and $|\phi_j|^2$: always positive, and independent of the relative spin of the two electrons.

Exchange integral:

$$K_{ij} = (ij|ji) = \iint \phi_i(1)\phi_j(1) (1/r_{12}) \phi_j(2)\phi_i(2) d\tau$$

This term has no classical counterpart - it comes purely from the antisymmetry of the Slater determinant. It is non-zero only between same-spin electrons and enters the energy with a minus sign, which is why same-spin (triplet) configurations sit lower than opposite-spin ones: the quantum origin of Hund's rule.

Exercise 6 - Excited states and spectra of CFH₂-OH (fluoromethanol)

Comparison table

State	TDA E / eV	TDA λ / nm	TDA f	CIS E / eV	CIS λ / nm	CIS f
S1	8.10	153.1	0.0087	9.81	126.4	0.0082
S2	8.90	139.2	0.0193	11.05	112.2	0.0444
S3	9.43	131.5	0.0456	11.86	104.5	0.1865
S4	9.93	124.8	0.0144	12.43	99.7	0.0165
S5	10.10	122.8	0.0586	13.03	95.2	0.2335

UV/Vis spectrum

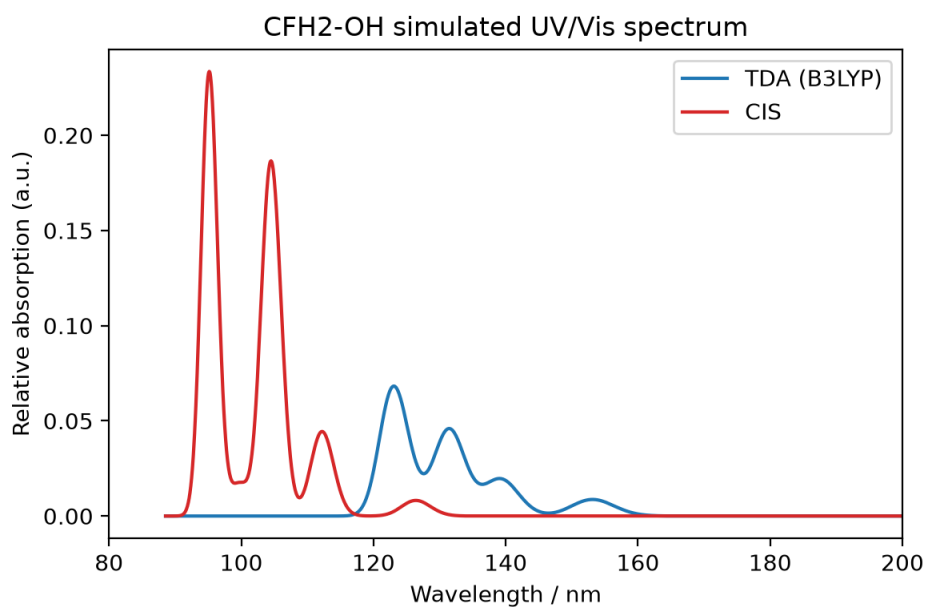


Figure 1. Simulated UV/Vis spectrum (Gaussian broadening, FWHM = 0.4 eV) from the TDA and CIS excitation energies and oscillator strengths above.

CIS gives systematically higher excitation energies than TDA/B3LYP across all five states (by roughly 1.7-3.0 eV), consistent with CIS lacking electron correlation.